

Inductive spatial transfer: experiments

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1 Design

- Train on one region, predict a second region the model has never seen.
- The two regions carry separate graphs. No edge crosses the boundary.
- Inductive: the test graph did not exist at training time.
- Metric is $R^2 = 1 - \text{SSE}/\text{SST}$ on the held-out region. AUC for classification.
- Arms: OLS, spatial error model (GMM), XGBoost, XGBoost with row-standardised spatial lags of every covariate, GraphSAGE, GraphSAGE with node-level layer normalisation.
- Stochastic arms run 10 to 15 seeds; deterministic arms run once.

2 Held-out US states

3,005 counties in the 39 states holding at least 20 counties. Target is county two-party Democratic vote share from 14 demographic and economic covariates. Hold out one state, train on the other 38, predict every county in the held-out state.

Table 1: Mean over the 39 held-out states.

Model	R2	RMSE	MAE
GraphSAGE	0.645	7.041	5.565
GraphSAGE + LayerNorm	0.617	7.332	5.729
XGBoost + lags	0.542	8.061	6.357
XGBoost	0.532	8.124	6.385
OLS	0.187	10.047	7.946

Model	R2	RMSE	MAE
SEM	0.186	10.246	8.332

Table 2: Paired over states. Percentages are means of per-state change.

Comparison	R2 difference	p value	States lower error	RMSE pct	MAE pct
GraphSAGE less XGBoost + lags	0.103	0.001	30/39	11.944	11.442

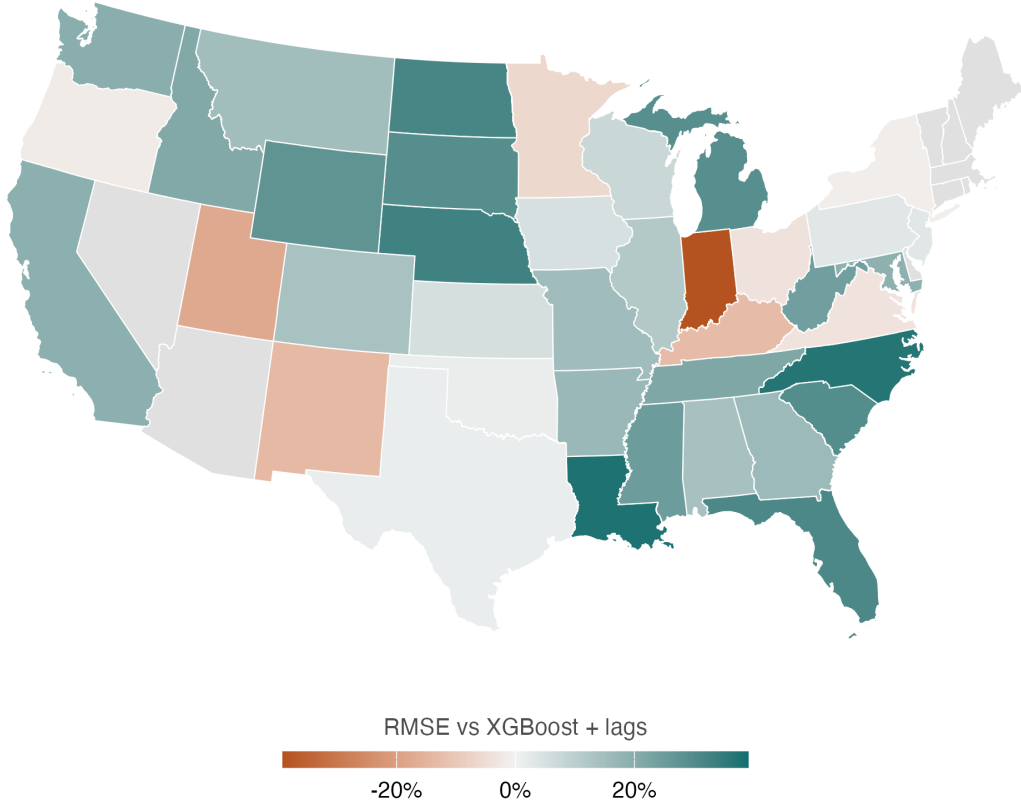


Figure 1: Per-state change in RMSE, GraphSAGE against XGBoost with spatial lags. Grey states hold fewer than 20 counties and were not tested.

3 Cross-region transfers

Margin is the best GraphSAGE variant less the best arm that uses no graph, both on the held-out region.

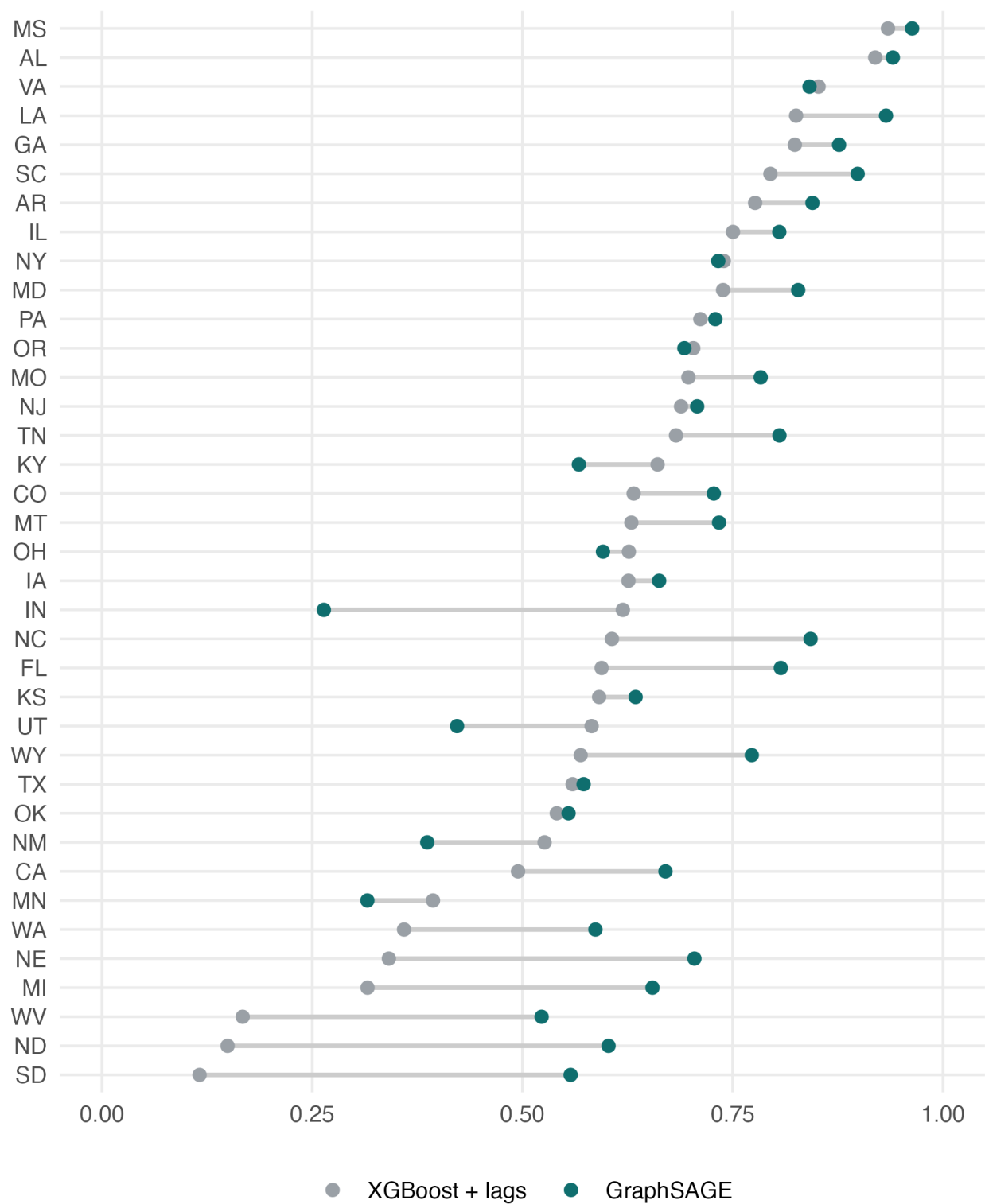


Figure 2: R-squared on each held-out state, ordered by the XGBoost result. Wisconsin and Idaho are below zero for both models and are omitted from the axis.

Table 3: Ordered by margin.

Transfer	GraphSAGE	Best other	Arm	Margin
County vote share, 39 held-out states	0.645	0.542	XGBoost + lags	0.103
Surface temperature, Cedar Rapids to Des Moines	0.796	0.699	XGBoost + lags	0.097
Airbnb price, LA to Chicago	0.630	0.588	XGBoost + lags	0.042
House price, Ames to King County	0.307	0.267	OLS	0.040
Airbnb price, Broward to San Diego	0.679	0.649	XGBoost	0.030
Surface temperature, Des Moines to Cedar Rapids	0.817	0.808	XGBoost + lags	0.009
Airbnb price, Chicago to LA	0.623	0.632	OLS	-0.009
311 response time, Queens to Brooklyn	0.254	0.301	OLS	-0.047
311 response time, Brooklyn to Queens	0.446	0.494	XGBoost	-0.048
Airbnb price, San Diego to Broward	0.484	0.540	XGBoost	-0.055
Diabetes prevalence, CA to WV	0.741	0.801	XGBoost + lags	-0.060
Airbnb reviews, Broward to San Diego	0.198	0.271	XGBoost	-0.074
Tree diameter, Brooklyn to Queens	0.246	0.334	XGBoost	-0.088
Tree diameter, Queens to Brooklyn	0.167	0.275	OLS	-0.109

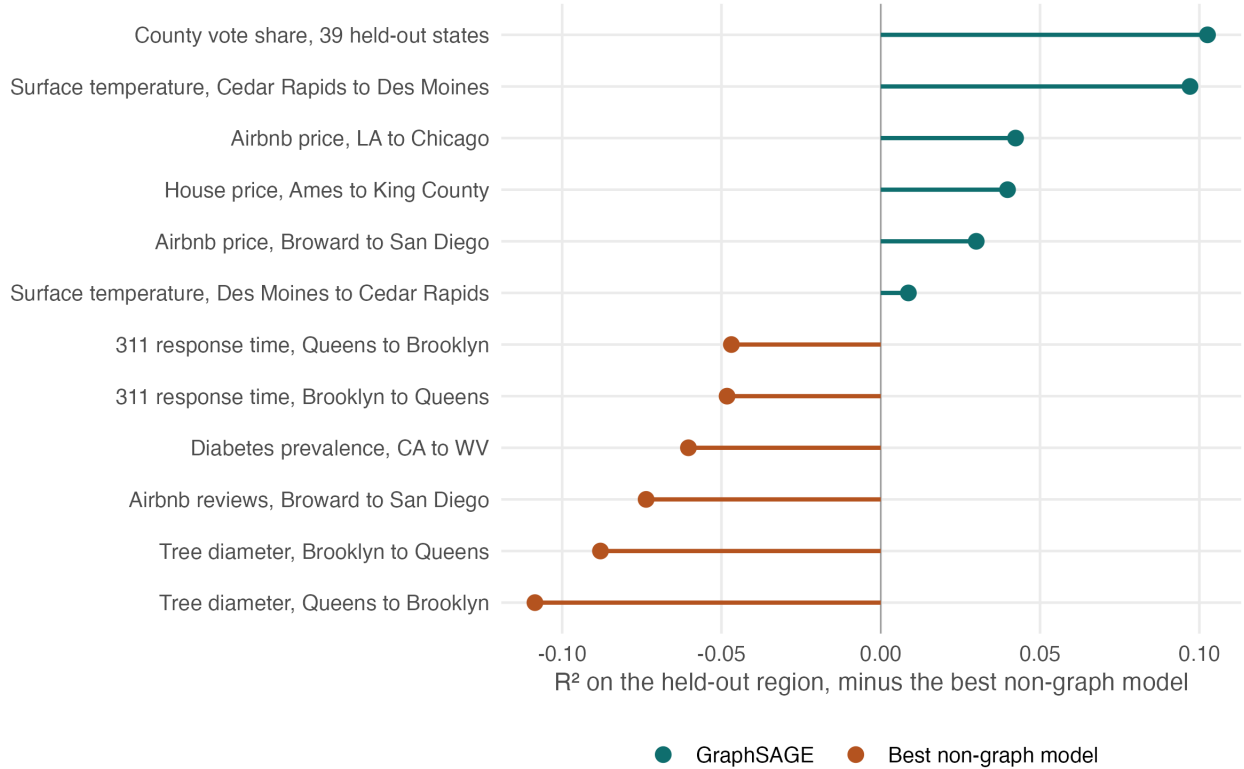


Figure 3: Margin over the best arm without a graph. Land cover is scored by AUC and is excluded from this axis.

3.1 Classification

Same two cities and the same predictors as the surface temperature transfer, with land cover class as the target.

Table 4: Des Moines to Cedar Rapids, land cover class.

Model	AUC
GraphSAGE + LayerNorm	0.9277
Logistic	0.9267
GraphSAGE	0.9191
XGBoost + lags	0.8893
XGBoost	0.8775

Land cover is more spatially autocorrelated than surface temperature. GraphSAGE gains 0.097 R^2 on temperature and 0.001 AUC on land cover, so autocorrelation of the target does not order the two outcomes.

4 Layer normalisation

Node-level layer normalisation against plain GraphSAGE, paired by seed. SD ratio is the median ratio of feature standard deviations between the two regions, computed from covariates only.

Table 5: Ordered by difference.

Transfer	GraphSAGE	With LayerNorm	Difference	SD ratio
House price, King County to Ames	-1.383	0.149	1.531	1.385
Diabetes prevalence, CA to WV	0.166	0.741	0.575	1.569
House price, Ames to King County	0.037	0.290	0.253	1.385
Tree diameter, Queens to Brooklyn	0.028	0.149	0.121	1.200
Airbnb price, Chicago to LA	0.550	0.620	0.069	1.082
Airbnb reviews, Broward to San Diego	0.125	0.194	0.069	1.077
Surface temperature, Des Moines to Cedar Rapids	0.755	0.822	0.067	1.063
Airbnb price, San Diego to Broward	0.404	0.449	0.045	1.077
Tree diameter, Brooklyn to Queens	0.197	0.237	0.040	1.200
Airbnb price, Broward to San Diego	0.673	0.679	0.007	1.077
311 response time, Brooklyn to Queens	0.439	0.438	-0.002	1.127
Airbnb price, LA to Chicago	0.630	0.623	-0.007	1.082
311 response time, Queens to Brooklyn	0.254	0.237	-0.016	1.127
Surface temperature, Cedar Rapids to Des Moines	0.796	0.778	-0.018	1.063
County vote share, 39 held-out states	0.645	0.617	-0.027	NA

Table 6: Split at 1.20. The cut is fitted to these transfers and is not validated out of sample.

Group	n	Difference above zero	Median	Min	Max
SD ratio at or above 1.20	3	3	0.575	0.253	1.531
SD ratio below 1.20	11	7	0.040	-0.018	0.121

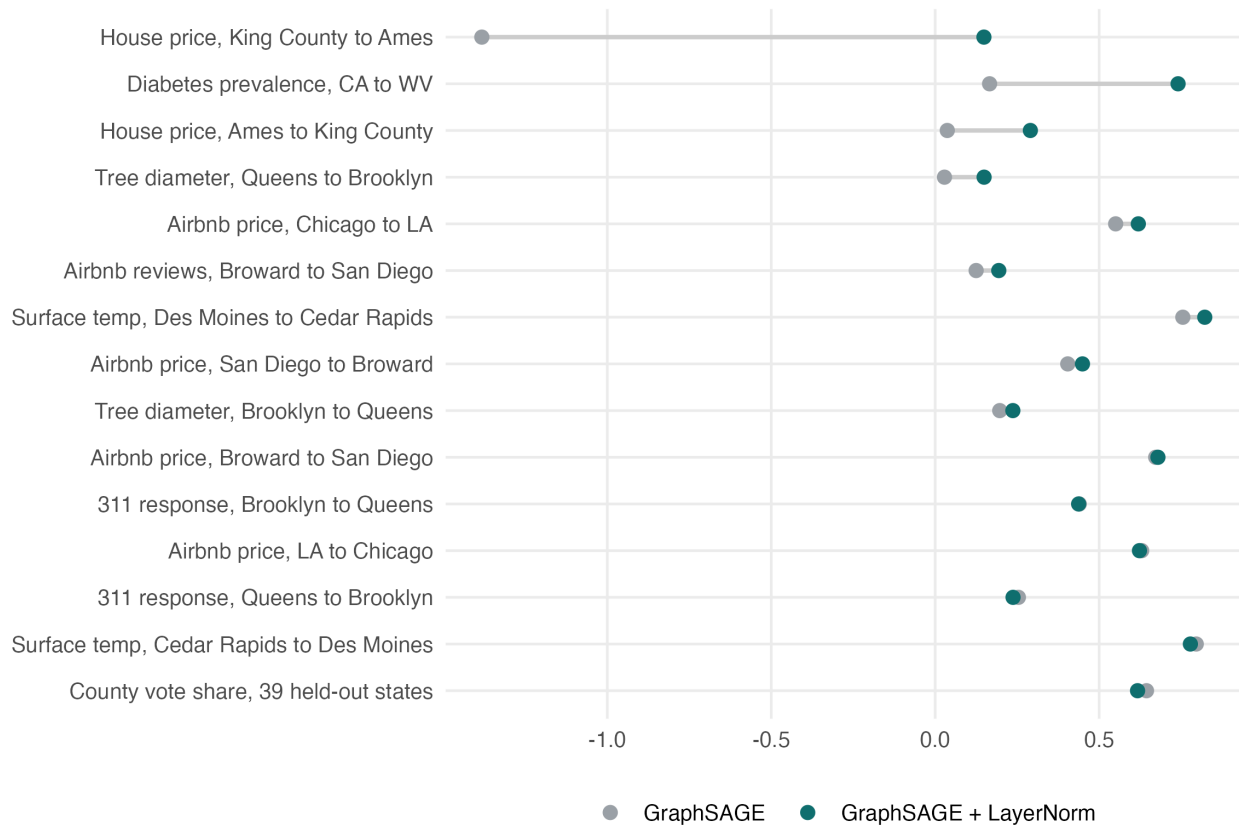


Figure 4: R-squared on the held-out region for both GraphSAGE variants.

Correlation of the difference with the SD ratio: 0.67 Pearson, 0.58 Spearman (n = 14).

The three largest differences occur where plain GraphSAGE returns a negative or near-zero R^2 . The SD ratio is computable from covariates before any target label is observed.

5 Findings

- GraphSAGE leads on 6 of 13 regression transfers and on the 39-state holdout. It trails on 6 and ties on the classification.
- The transfers it leads are surface temperature, vote share, and property price. The transfers it trails are tree diameter, 311 response time, review counts, and diabetes prevalence.
- Target autocorrelation does not order the outcomes. Land cover is more autocorrelated than surface temperature and shows the smaller margin.
- Direction is not symmetric. Broward to San Diego is +0.030 and San Diego to Broward is -0.055. Both surface temperature directions favour GraphSAGE; the margins differ by a factor of ten.
- Layer normalisation changes R^2 by at most 0.027 in either direction on 11 of 15 transfers. On the 4 with an SD ratio at or above 1.20 it raises R^2 by 0.12 to 1.53.

6 Not established

- Whether a transfer will succeed cannot be determined from the source region alone. Three pre-fit screens were tested against ten transfers with known outcomes; the wins and the losses overlap on all three.
- The held-out state results use one seed per state. A five-seed replication over all 39 states places GraphSAGE between 0.620 and 0.629; the other arms were fit once each.
- Neighbourhood size selected on held-out source data transfers poorly. Bandwidth is an absolute distance and the regions differ in point density. Selection by neighbour count is untested.
- The King County to Ames result reported previously at R^2 0.501 against 0.283 cannot be reproduced from `data/results/folds.parquet`: the 0.501 run carries no arm labels and no baseline at 0.283 appears in the data. It is excluded here.